JIRA FUNDAMENTALS AND HANDS-ON PRACTICE – SHOPEASY PROJECT

1. Introduction to Jira

1.1 What is Jira?

Jira is a project management and issue-tracking tool developed by Atlassian. It is widely used by software development teams to plan, track, manage, and complete project activities.

It helps teams organize requirements, assign tasks, track development progress, manage bugs, and collaborate efficiently.

1.2 Why Jira is Used

Jira helps teams to:

- Manage project requirements and tasks.
- Track software development progress.
- Organize User Stories and Epics.
- Identify and manage bugs.
- Plan and manage Sprints.
- Assign work to team members.
- Monitor project performance.
- Improve team collaboration.

2. Understanding Agile Methodology

2.1 What is Agile?

Agile is a software development methodology in which a project is divided into smaller parts and developed step by step.

Teams work in short development cycles called Sprints and continuously improve the product based on feedback.

2.2 Scrum Framework

Scrum is an Agile framework used to organize and manage software development activities.

Important Scrum concepts:

- Product Backlog
- Sprint
- Sprint Planning
- Daily Scrum
- Sprint Review
- Sprint Retrospective

2.3 Scrum Roles

- Product Owner: Manages project requirements and prioritizes the Product Backlog.
- Scrum Master: Helps the team follow Scrum practices and removes obstacles.
- Development Team: Develops and tests the software.

3. Creating a Scrum Project

Created a Scrum project in Jira for the ShopEasy e-commerce application.

Project Details

- Project Name: ShopEasy
- Project Type: Scrum
- Project Purpose: To organize and manage e-commerce application development using Agile methodology.
- Project Key: SHOP

Learned how to create a Scrum project, configure its settings, and understand the Jira project interface.

4. Understanding Epics

4.1 What is an Epic?

An Epic is a large piece of work that can be divided into multiple related User Stories.

It helps organize major functionalities of a project.

4.2 Creating Epics in ShopEasy

Created the following Epics to organize the major functionalities of the ShopEasy application.

Epic 1: User Management

Objective: Manage customer registration, login, and account details.

Epic 2: Product Management

Objective: Manage product information, categories, searching, and viewing products.

Epic 3: Shopping Cart

Objective: Allow customers to add, update, and remove products from their shopping cart.

Epic 4: Order Management

Objective: Manage order placement, order details, and order tracking.

Epic 5: Payment Management

Objective: Handle payment processing and payment confirmation.

Epics helped organize related User Stories under different functional modules.

5. Creating User Stories

5.1 What is a User Story?

A User Story describes a software feature from the user's point of view.

User Story Format:

As a user, I want to perform an action so that I can achieve a benefit.

5.2 User Stories in ShopEasy

Created User Stories for all major functionalities of the ShopEasy application.

Epic 1: User Management

1. As a user, I want to register an account so that I can access the ShopEasy application.

2. As a user, I want to log in to my account so that I can access personalized services.
3. As a user, I want to reset my password so that I can recover my account.
4. As a user, I want to update my profile so that my personal information remains current.

Epic 2: Product Management

5. As a user, I want to view available products so that I can choose products to purchase.
6. As a user, I want to search for products by name so that I can find products easily.
7. As a user, I want to filter products by category and price so that I can find suitable products.
8. As a user, I want to view product details so that I can make informed purchasing decisions.

Epic 3: Shopping Cart

9. As a user, I want to add products to my cart so that I can purchase them later.
10. As a user, I want to update product quantities so that I can select the required quantity.
11. As a user, I want to remove products from my cart so that I can manage my selections.
12. As a user, I want to view the total cart price so that I know the amount payable.

Epic 4: Order Management

13. As a user, I want to place an order so that I can purchase selected products.
14. As a user, I want to view my order history so that I can track previous purchases.
15. As a user, I want to track my order status so that I know the delivery progress.
16. As a user, I want to cancel an eligible order so that I can stop an unwanted purchase.

Epic 5: Payment Management

17. As a user, I want to select a payment method so that I can pay conveniently.
18. As a user, I want to complete an online payment so that my order can be confirmed.
19. As a user, I want to receive payment confirmation so that I know whether my transaction was successful.
20. As a user, I want to view payment details so that I can verify my transactions.

6. Understanding Acceptance Criteria

6.1 What are Acceptance Criteria?

Acceptance Criteria define the conditions that must be satisfied for a User Story to be considered complete.

They help developers and testers understand the expected functionality.

6.2 Acceptance Criteria for ShopEasy User Stories

US-01: User Registration

- User must enter a valid name, email, and password.
- All required fields must be filled.
- Duplicate email registration must be prevented.
- Successful registration must create a user account.
- Appropriate validation messages must be displayed.

US-02: User Login

- User must enter registered email and password.
- Valid credentials must allow login.
- Invalid credentials must display an error message.
- Empty fields must not be accepted.

US-03: Password Reset

- User must enter a registered email address.
- The system must initiate password recovery.
- Invalid email addresses must display an appropriate message.

US-04: Update Profile

- User must be able to edit profile details.
- Updated information must be saved.
- Invalid information must be rejected.

US-05: View Products

- Available products must be displayed.
- Product names, prices, and images must be visible.
- Unavailable products must be identified.

US-06: Search Products

- User must be able to search using product names.
- Matching products must be displayed.
- No matching products must show an appropriate message.

US-07: Filter Products

- User must be able to filter by category and price.
- Only matching products must be displayed.
- Filters must be removable.

US-08: View Product Details

- Product name, price, image, and description must be displayed.
- Product availability must be visible.

US-09: Add Products to Cart

- User must be able to add an available product.
- Selected product must appear in the cart.
- Cart quantity must be updated correctly.

US-10: Update Cart Quantity

- User must be able to increase or decrease quantity.
- Total price must be recalculated.
- Invalid quantities must not be accepted.

US-11: Remove Products from Cart

- User must be able to remove selected products.
- Removed products must disappear from the cart.
- Cart total must be updated.

US-12: View Cart Total

- All selected product prices must be calculated.
- Quantity must be considered.
- Correct total amount must be displayed.

US-13: Place Order

- User must have products in the cart.
- Required delivery details must be provided.
- Order must be created after successful confirmation.

US-14: View Order History

- Previous orders must be displayed.
- Order details and dates must be visible.
- Users must see only their own orders.

US-15: Track Order

- Current order status must be displayed.
- Order progress must be updated.
- Invalid order references must be handled.

US-16: Cancel Order

- Eligible orders must be cancellable.
- Cancelled orders must show the updated status.
- Completed or otherwise ineligible orders must not be cancelled.

US-17: Select Payment Method

- Available payment methods must be displayed.
- User must select a valid payment method.
- Payment details must be validated.

US-18: Complete Payment

- Payment must be processed securely.
- Successful payment must be recorded.
- Failed payment must display an error.
- An order must not be marked paid after a failed transaction.

US-19: Payment Confirmation

- Successful transactions must generate confirmation.
- Transaction reference must be available.
- Failed transactions must show the appropriate status.

US-20: View Payment Details

- User must be able to view transaction details.
- Payment amount and status must be displayed.
- Payment information must be associated with the correct order.

7. Understanding Jira Work Types

Learned about different work types used in Jira.

- Epic: A large functionality or requirement.
- Story: A feature requested by the user.
- Task: A specific piece of work.
- Bug: An issue or defect in the application.
- Sub-task: A smaller activity created under a parent task.
- Test: Used to represent a test case when supported by the testing setup.

These work types help teams organize, assign, and track project activities.

8. Creating Tasks and Sub-tasks

8.1 What is a Task?

A Task represents a specific activity that needs to be completed during project development.

8.2 Tasks Created in ShopEasy

Created the following development tasks:

1. Design the user registration page.
2. Implement user login functionality.
3. Develop password reset functionality.
4. Create the product listing page.
5. Implement product search functionality.
6. Develop product filtering.
7. Design the shopping cart interface.
8. Implement cart quantity updates.
9. Develop order placement functionality.
10. Integrate payment processing.
11. Implement order tracking.
12. Create order history functionality.

8.3 What is a Sub-task?

A Sub-task is a smaller activity created under a parent Task or another supported parent issue.

Examples:

Task: Implement User Registration

- Design registration form.
- Validate user inputs.
- Store user information.
- Test registration functionality.

Task: Implement Shopping Cart

- Create cart interface.
- Add products to cart.
- Update product quantities.
- Calculate total price.

Sub-tasks helped divide large activities into smaller manageable tasks.

9. Sprint Planning and Backlog Management

9.1 What is a Product Backlog?

A Product Backlog is a prioritized list of all requirements and features that need to be developed in a project.

9.2 What is Sprint Planning?

Sprint Planning is a meeting in which the team selects backlog items to complete during a Sprint.

9.3 Sprint Planning in ShopEasy

Organized the ShopEasy project into the following development Sprints.

Sprint 1: User Management

- User registration.
- User login.
- Password reset.
- Profile management.

Sprint 2: Product Management

- View products.
- Search products.
- Filter products.
- View product details.

Sprint 3: Shopping Cart

- Add products to cart.
- Update quantities.
- Remove products.
- Calculate cart total.

Sprint 4: Order and Payment Management

- Place orders.
- View order history.
- Track orders.
- Cancel orders.
- Select payment methods.
- Process payments.
- Generate payment confirmations.

Learned how to create Sprints, add backlog items, start Sprints, and monitor Sprint progress.

10. Understanding Story Points and Estimation

10.1 What are Story Points?

Story Points are used to estimate the relative effort, complexity, and amount of work required to complete a User Story.

10.2 Story Point Estimation

Used the Fibonacci sequence for estimating User Stories.

Common Story Point values:

1, 2, 3, 5, 8, 13

Example estimation:

User Story	Story Points
User Registration	3
User Login	2
Password Reset	3
View Products	3
Search Products	5
Filter Products	5
Add Products to Cart	3
Update Cart Quantity	2
Place Order	8
Payment Processing	8

Learned how to estimate work complexity and use Story Points during Sprint Planning.

11. Understanding Test Cases

11.1 What is a Test Case?

A Test Case is a set of steps, inputs, and expected results used to verify whether a particular functionality works correctly.

Test Cases help identify defects and ensure that the application behaves as expected.

11.2 Test Cases for ShopEasy

Created and organized test cases for the major functionalities of ShopEasy.

Test Case ID	Test Scenario	Expected Result
TC-01	Register with valid details	Account created successfully
TC-02	Register with an existing email	Duplicate registration rejected
TC-03	Register with empty fields	Validation message displayed
TC-04	Login with valid credentials	User logged in successfully
TC-05	Login with invalid password	Error message displayed
TC-06	Reset password with registered email	Password recovery initiated

Test Case ID	Test Scenario	Expected Result
TC-07	Update profile information	Profile updated successfully
TC-08	View available products	Products displayed correctly
TC-09	Search for an existing product	Matching products displayed
TC-10	Search for a nonexistent product	No products found message displayed
TC-11	Filter products by category	Matching category products displayed
TC-12	View product details	Correct product information displayed
TC-13	Add product to cart	Product added successfully
TC-14	Increase product quantity	Quantity and total updated
TC-15	Remove product from cart	Product removed successfully
TC-16	View cart total	Correct total amount displayed
TC-17	Place order with valid details	Order created successfully
TC-18	View order history	Previous orders displayed
TC-19	Track an existing order	Correct order status displayed
TC-20	Cancel an eligible order	Order status changed to Cancelled
TC-21	Select payment method	Payment method selected successfully
TC-22	Complete successful payment	Payment recorded successfully
TC-23	Complete failed payment	Payment failure message displayed
TC-24	View payment confirmation	Transaction confirmation displayed
TC-25	View payment details	Correct transaction details displayed

11.3 Test Case Execution

Performed test case execution by comparing actual results with expected results.

Test cases were organized based on their related application functionalities.

The testing process helped verify requirements and identify defects.

12. Bug Reporting and Defect Management

12.1 What is a Bug?

A Bug is an error or defect in an application that causes it to behave differently from the expected result.

12.2 Creating and Linking Bugs

Learned how to create Bugs in Jira and associate them with related User Stories and testing activities.

A Bug generally contains:

- Bug Summary.
- Description.
- Steps to Reproduce.
- Expected Result.

- Actual Result.
- Priority.
- Severity.
- Environment.
- Related User Story or Test Case.

12.3 Sample Bugs Identified in ShopEasy

BUG-01: Login with Invalid Password

- Description: Application does not display the expected error message for invalid login credentials.
- Expected Result: Login should be rejected with an error message.
- Actual Result: Incorrect login response is displayed.
- Priority: High.

BUG-02: Incorrect Cart Total

- Description: Cart total is not updated after changing product quantity.
- Expected Result: Total amount should be recalculated.
- Actual Result: Previous total amount remains displayed.
- Priority: High.

BUG-03: Product Search Not Working

- Description: Searching for an available product does not display matching results.
- Expected Result: Matching products should be displayed.
- Actual Result: No results are displayed.
- Priority: Medium.

BUG-04: Duplicate Order Creation

- Description: Repeated order submission creates duplicate orders.
- Expected Result: Only one order should be created.
- Actual Result: Multiple orders are generated.
- Priority: High.

BUG-05: Payment Confirmation Missing

- Description: Successful payment does not display the confirmation message.
- Expected Result: Payment confirmation should be displayed.
- Actual Result: Confirmation message is missing.
- Priority: High.

12.4 Bug Life Cycle

Understood the common Bug Life Cycle:

1. Open – Bug is reported.
2. In Progress – Developer starts fixing the bug.
3. Fixed – Developer completes the fix.
4. Retest – Tester verifies the fix.
5. Closed – Bug is confirmed as resolved.

This process helps track defects until they are resolved.

13. Understanding Test Plans

13.1 What is a Test Plan?

A Test Plan is a document used to organize testing activities for a project or software release.

It defines testing objectives, scope, resources, test cases, and testing activities.

13.2 Test Plan for ShopEasy

Test Plan Name: ShopEasy Functional Testing Plan

Objective: To verify that all major ShopEasy application functionalities work correctly.

Testing Scope:

- User registration and login.
- Product management.
- Shopping cart.
- Order management.
- Payment management.

Testing Activities:

1. Identify application requirements.
2. Prepare test cases.
3. Execute test cases.
4. Compare actual and expected results.
5. Report defects.
6. Retest fixed defects.
7. Complete testing and report results.

Learned how test plans help organize test cases and monitor testing activities.

14. Understanding Jira Workflows

14.1 What is a Workflow?

A Workflow represents the different stages through which a Jira issue moves from creation to completion.

14.2 Common Workflow Statuses

- Open: Issue is newly created.
- To Do: Work is waiting to be started.
- In Progress: Work is currently being performed.
- Done: Work has been completed.

14.3 Workflow in ShopEasy

The development workflow was organized as:

To Do -> In Progress -> Done

For Bugs, the testing and defect-resolution process was understood as:

Open -> In Progress -> Fixed -> Retest -> Closed

Learned how to update issue statuses and track development progress using Jira workflows.

15. Jira Dashboards

15.1 What is a Dashboard?

A Jira Dashboard provides a summary of project information in one place.

It helps teams monitor project progress, issue status, Sprint performance, and other important project information.

15.2 Dashboard Components

Explored the following dashboard gadgets:

- Filter Results: Displays issues based on selected filters.
- Pie Chart: Displays issues grouped by status or priority.
- Created vs Resolved Chart: Shows the number of issues created and resolved.
- Sprint Burndown Chart: Tracks remaining work during a Sprint.
- Assigned to Me: Displays issues assigned to the current user.
- Activity Stream: Displays recent project activities.

Learned how dashboards help monitor ShopEasy project progress.

16. Understanding JQL Filters

16.1 What is JQL?

JQL stands for Jira Query Language.

It is used to search and filter Jira issues based on specific conditions such as project, issue type, status, priority, and assignee.

16.2 Common JQL Queries

1. Display all issues in ShopEasy

```
project = SHOP
```

2. Display all Stories

```
project = SHOP AND issuetype = Story
```

3. Display all Bugs

```
project = SHOP AND issuetype = Bug
```

4. Display issues currently in progress

```
project = SHOP AND status = "In Progress"
```

5. Display issues assigned to the current user

```
project = SHOP AND assignee = currentUser()
```

6. Display high-priority issues

```
project = SHOP AND priority = High
```

7. Display completed issues

```
project = SHOP AND status = Done
```

8. Display issues belonging to a particular Sprint

```
project = SHOP AND Sprint = 1
```

These queries helped retrieve specific issues and monitor project activities efficiently.

17. Hands-On Practice and Learning

During the practical session, I performed the following activities using Jira:

1. Created a Scrum project named ShopEasy.
2. Explored the Jira project interface.
3. Created and organized five Epics.
4. Created 20 User Stories.
5. Added Acceptance Criteria to User Stories.
6. Created Tasks and Sub-tasks.
7. Explored different Jira Work Types.
8. Organized the Product Backlog.
9. Planned four development Sprints.
10. Estimated User Stories using Story Points.
11. Created and organized 25 Test Cases.
12. Understood Test Case execution.
13. Created and linked sample Bugs.
14. Explored the Bug Life Cycle.
15. Understood Test Plans.
16. Configured and explored Jira Workflows.
17. Explored Jira Dashboards and Gadgets.
18. Created and executed JQL filters.
19. Monitored project progress using Jira boards.
20. Understood the complete Agile Scrum development process.

18. Complete Jira Project Workflow

Learned the overall project management workflow using Jira for the ShopEasy application.

1. Create a Scrum Project.
2. Identify project requirements.
3. Create Epics.
4. Create User Stories under Epics.
5. Define Acceptance Criteria.
6. Create Tasks and Sub-tasks.
7. Organize work in the Product Backlog.
8. Plan development Sprints.
9. Estimate work using Story Points.
10. Start and manage Sprints.
11. Track issues using Workflows.
12. Prepare and execute Test Cases.
13. Create and manage Bugs.
14. Retest fixed defects.
15. Monitor project progress using Dashboards.
16. Search and filter issues using JQL.
17. Review completed work and conduct Sprint Retrospectives.

19. Conclusion

Through this practical training, I gained hands-on knowledge of Jira and Agile project management using the ShopEasy e-commerce project.

I learned how to create and manage Scrum projects, organize Epics and User Stories, define Acceptance Criteria, create Tasks and Sub-tasks, plan Sprints, and estimate work using Story Points.

I also gained knowledge of Test Cases, Test Plans, Bug Reporting, Bug Life Cycle, Jira Workflows, Dashboards, and JQL queries.

This practical experience helped me understand how Jira is used in real-world software development projects to manage requirements, coordinate team activities, track development progress, and maintain software quality.